Reviewer(s)' Comments to Author:

Link to document for revisions

Reviewer: 1

Comments to the Author

This manuscript provides a comprehensive overview of the accomplishments and activities of the AgBioData Consortium, including the development of community structures, training initiatives, interoperability efforts, and FAIR data resources. Overall, I am very impressed by the scope and ambition of the work described.

Several aspects of the project are particularly noteworthy. The funding mechanisms supporting working-group leadership, conference participation, and curriculum implementation appear exceptionally valuable for fostering sustained community engagement. The training resources and FAIR-focused educational materials are detailed, thoughtfully designed, and clearly represent a substantial contribution to the community. Likewise, the database finder resource appears highly useful for helping researchers identify appropriate repositories and resources.

More broadly, the manuscript is well written and succeeds in documenting the extensive efforts undertaken by the consortium over the past several years. The underlying work itself is clearly impactful and deserving of recognition.

At the same time, I remain uncertain whether the manuscript, in its current form, provides sufficient novel actionable insight to warrant publication in Database. In its present structure, the paper reads more as a synthesis or programmatic overview of accomplishments than as a manuscript centered on novel methodological, infrastructural, or empirical contributions. This concern is not a reflection of the value of the consortium's work, which I view very positively, but rather of the fit between the manuscript's current framing and the journal's typical scope.

Kind regards,
Sven Lesche

P.S. An AI-based language tool was used solely for editing and phrasing; all evaluations, interpretations, and substantive content are entirely my own.

Major Comments

First, I want to clarify that I am not a specialist in agricultural biology research specifically; my background is in psychology and FAIR/open-data practices more broadly. As such, I will focus this review primarily on the manuscript's contribution to data infrastructure, interoperability, and FAIR implementation.

1. The manuscript's novel contribution to the journal remains unclear

As stated on p. 4 (ll. 113–116), the primary goal of the manuscript is:

> "to provide an overview of the efforts and accomplishments [...] of AgBioData, [...] their impact on awareness and implementation of FAIR principles [...], and [...] future directions."

This framing creates some uncertainty regarding the manuscript's primary scholarly contribution. A descriptive overview of accomplishments can certainly be valuable, but in its current form the manuscript often reads more like a consortium report or retrospective summary than a paper advancing a clearly novel set of findings, methods, or recommendations.

The manuscript does summarize a number of important outputs from the working groups (including ontology recommendations, SNP harmonization protocols, interoperability initiatives, and FAIR training curricula) but these contributions appear to have already been published in their own dedicated papers. In the current manuscript, they are largely summarized rather than extended.

This creates some ambiguity regarding what specifically is new in the present article beyond the aggregation of prior accomplishments. For example, the **abstract** highlights:

> "Major outcomes include guidelines for standardizing genetic and genomic data, SNP harmonization protocols, recommendations for ontology usage, and an open-access curriculum for FAIR data literacy."

However, these outputs themselves do not appear to be introduced here for the first time.

Similarly, the statement that:

> "AgBioData represents a scalable model for enhancing agricultural data infrastructure and sustainability"

could form the basis of a strong contribution, but the manuscript currently places greater emphasis on recounting activities and accomplishments than on extracting generalizable principles or actionable guidance for other communities seeking to build similar infrastructures.

To reiterate, I view the underlying work of the consortium very positively and greatly appreciate the extensive effort reflected throughout the manuscript. My concern is specifically whether the current paper provides sufficient novel insight, synthesis, or actionable guidance (beyond documenting prior activities) to justify publication in its current form.

Response: We thank the reviewer for the generous assessment of the consortium's work and for articulating this concern so clearly. We agree that our novel contribution should be stated more explicitly, and we have sharpened the framing throughout rather than restructuring the paper. Our intended contribution is not the individual working group outputs (each published or being published, separately), rather providing a generalized operating model for building FAIR data infrastructure within a research domain especially the participatory bottom up process by which data priorities are identified, working groups are scoped and run, contributors are incentivized, and impact is evaluated through repeated stakeholder surveys. In addition, we have provided the cross-cutting, actionable lessons learned that other communities can reuse. We also note that several resources are introduced here for the first time, including the AgBioData Database Finder, the integrated incentive framework (the Champions, Ambassador, and Curriculum Implementation programs), and the 2022-2024 survey evidence of changing FAIR awareness and adoption.

Minor Comments

1. Evaluation of the awards and support programs

The various awards and support programs described throughout the manuscript appear highly valuable for fostering community engagement and FAIR adoption. However, the manuscript would benefit from a clearer evaluation of their uptake and impact. For example, are these programs widely used? Has participation increased over time? Are there any indicators that they have successfully improved engagement, training, or FAIR implementation? Even a brief summary of participation metrics or outcomes would strengthen the discussion of these initiatives.

Response: t the time the manuscript was prepared we had not yet launched the initiatives. We have now provided information about the participation metrics and outcomes.

2. Clarification regarding barriers to data deposition and curation

On p. 13 (ll. 397–410), the manuscript discusses bottlenecks in the publication–curation pipeline and provides useful recommendations regarding training and repository identification. The database finder resource also appears particularly valuable in this regard.

However, I was surprised that incentives and time costs associated with data submission and curation were not discussed as barriers. In my own experience with FAIR data initiatives, lack of incentives and the substantial time investment required for proper curation are often major obstacles to database contribution. If these issues did not emerge during the consortium's activities or surveys, that would itself be informative and worth stating explicitly. Alternatively, if they were identified as barriers, they would merit further discussion alongside potential mitigation strategies.

Response:

3. Declining reported usage of GGB databases in survey results

Table S2 reports stakeholder survey results regarding the use of GGB databases for data submission, searching, retrieval, and reanalysis. Across several categories, these **usage rates appear to decline rather than increase**. This seems potentially important, yet the manuscript does not substantially discuss this pattern.

The manuscript appropriately notes that the survey samples are not directly comparable, which may partially explain these findings. However, this same limitation would also apply to conclusions regarding increases in FAIR adoption. As currently presented, the interpretation of positive trends in FAIR adoption appears somewhat stronger than the interpretation of potentially negative trends regarding GGB resource usage.

Relatedly, Table S3 appears to show that increases in FAIR-related adoption exceed increases specifically associated with GGB-related resources and practices. This raises an interesting possibility that broader FAIR awareness may have grown more rapidly than direct engagement with consortium-specific infrastructure. If so, this would be valuable to discuss explicitly as part of the overall evaluation of the consortium's impact.

4. Sections with unclear relevance to the journal audience

Related to the major concern above, several sections of the manuscript read more like organizational reporting than **actionable guidance** for the broader research community.

For example, p. 17 (ll. 532–538) emphasizes the importance of the program coordinator role. While this may well have been essential to the consortium's success, the practical implication for readers is currently unclear. If the intent is to recommend that similar initiatives adopt dedicated coordination structures, this could be stated more explicitly and framed as a transferable organizational lesson.

Similarly, p. 18 (ll. 546–573) primarily outlines future activities and funding needs of the consortium. While understandable in the context of a long-running community initiative, this

section currently contains relatively limited actionable information for readers of Database. Reframing these sections toward generalizable recommendations or lessons learned would strengthen their relevance to the journal audience.

Response: We have tried to reframe this to include generalizable recommendations and lessons learned.

5. Readability of Table S1 in the PDF version

In the PDF version, some cells in Table S1 appear truncated, with text cut off at the margins. I recognize that this issue is not the case for the electronically shared ".xlsx" files and thus may ultimately be resolved during production or electronic publication. Insofar as the supplementary tables appear in printed form, it would be worth ensuring that the table is fully readable in the final version.

Response: Agreed and hopefully the revised PDF is fully readable.

6. Minor typographical and formatting issues

- p. 17, l. 529: there appear to be two consecutive punctuation marks.
- Table S2, row “Phenotypic Data Standardization and Management (PDSM),” column “Deliverables”: the manuscript title should be formatted in bold for consistency with the remainder of the table.

Response: Apologies. Both formatting issues should be resolved.

Reviewer: 2

Comments to the Author

Really cool project doing all the right things, well done. I just have two minor notes.

- Lines 477 to 480: "97% rated the application of community standards by publishers... as 'very important' or 'highest priority'." and "Professional incentives from funders were similarly rated..." I would like to hear more commentary about these topics because they were rated so highly by survey respondees. These topics address structural issues in the scientific process related to publishers and funders. Your insights would be valuable.

Response: We have added a little bit of commentary based on presentations from the FAIR literature working group.

- ❖ Also, in the final version I hope that the **three tables** are not presented sideways. Sideways figures are very inaccessible (and annoying).

Response: Hopefully we have addressed this concern regarding the supplemental files S2-5.